## 2.3 Rhombus

Take four sticks of equal length and four connectors. Connect four sticks to form a square as shown in the Fig.2.11(i). Then, try to make any two opposite vertices closer as shown in the Fig.2.11(ii) such that opposite sides remain parallel to each other to get a new shape called rhombus.

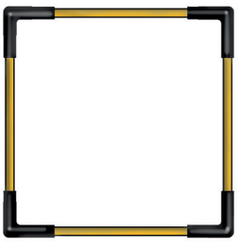


Fig.2.11(i)

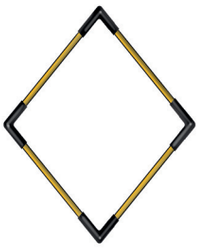


Fig.2.11(ii)

Hence, we conclude that, in a parallelogram, if all the sides are equal then it is called a rhombus.

**Note**

In a rhombus, (i) all the sides are equal (ii) opposite sides are parallel (iii) diagonals divide the rhombus into 4 right angled triangles of equal area. (iv) the diagonals bisect each other at right angles.

### 2.3.1 Area of the rhombus if base and height are given

Draw a rhombus on a graph sheet as shown in the Fig.2.12(i) and cut it. Draw a perpendicular line from one vertex to the opposite side. Cut the triangle and shift the triangle to the other side of the rhombus as shown in the Fig.2.12(ii).What shape do you see? It is a rectangle. Hence, the area of the rhombus is the same as that of the rectangle.

Area of the rectangle

= length × breadth $sq.units$

= base × height $sq.units$

= area of the rhombus

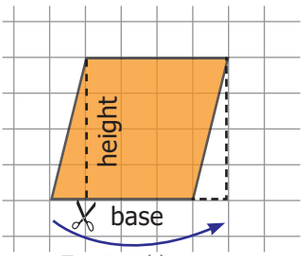



Fig.2.12(i)

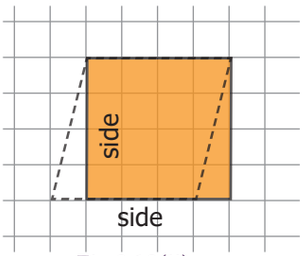



Fig.2.12(ii)

### 2.3.2 Area of the rhombus if the diagonals are given

Let us find, area of the rhombus ABCD by splitting it into two triangles.

Here AB = BC = CD = DA and diagonals AC ($d_1$) and BD ($d_2$) are perpendicular to each other.

Area of the rhombus ABCD = Area of triangle ABC + Area of triangle ADC

$$= \frac{1}{2} \times AC \times OB + \frac{1}{2} \times AC \times OD$$

$$= \frac{1}{2} \times AC\ (OB+OD)$$

$$= \frac{1}{2} \times AC \times BD$$

$$= \frac{1}{2} \times d_1 \times d_2\ sq.\ units$$

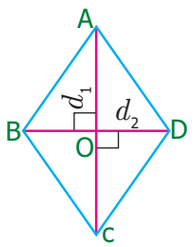



Fig.2.13

Therefore, area of the rhombus = $\frac{1}{2}$ (product of diagonals) $square\ units$.



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